

High-order interactions

High-order interactions are critical in molecular systems: conjugated π -systems, hydrogen bonding networks, and ring strain involve multiple atoms at once.

Traditional Message Passing Networks model only pairwise connections via edges. They miss multi-body dependencies and ignore 3D geometry

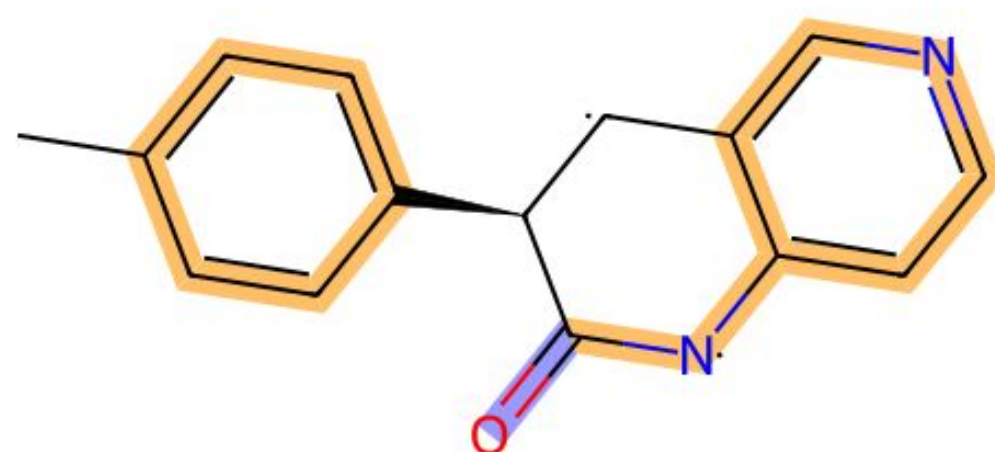


Figure 1: Hypergraph representation of a molecule, where nodes correspond to atoms and hyperedges represent conjugated bonds.

Hypergraphs solve this: hyperedges connect any number of atoms, naturally capturing topological high-order interactions.

Bipartite Representation

The AllSet^[1] framework models hypergraphs as bipartite graphs with two partitions - nodes (atoms) and hyperedges (chemical groups).

It performs message passing in two permutation-invariant phases: first, each hyperedge gathers information from its connected nodes; then, each node updates its state based on all hyperedges it participates in.

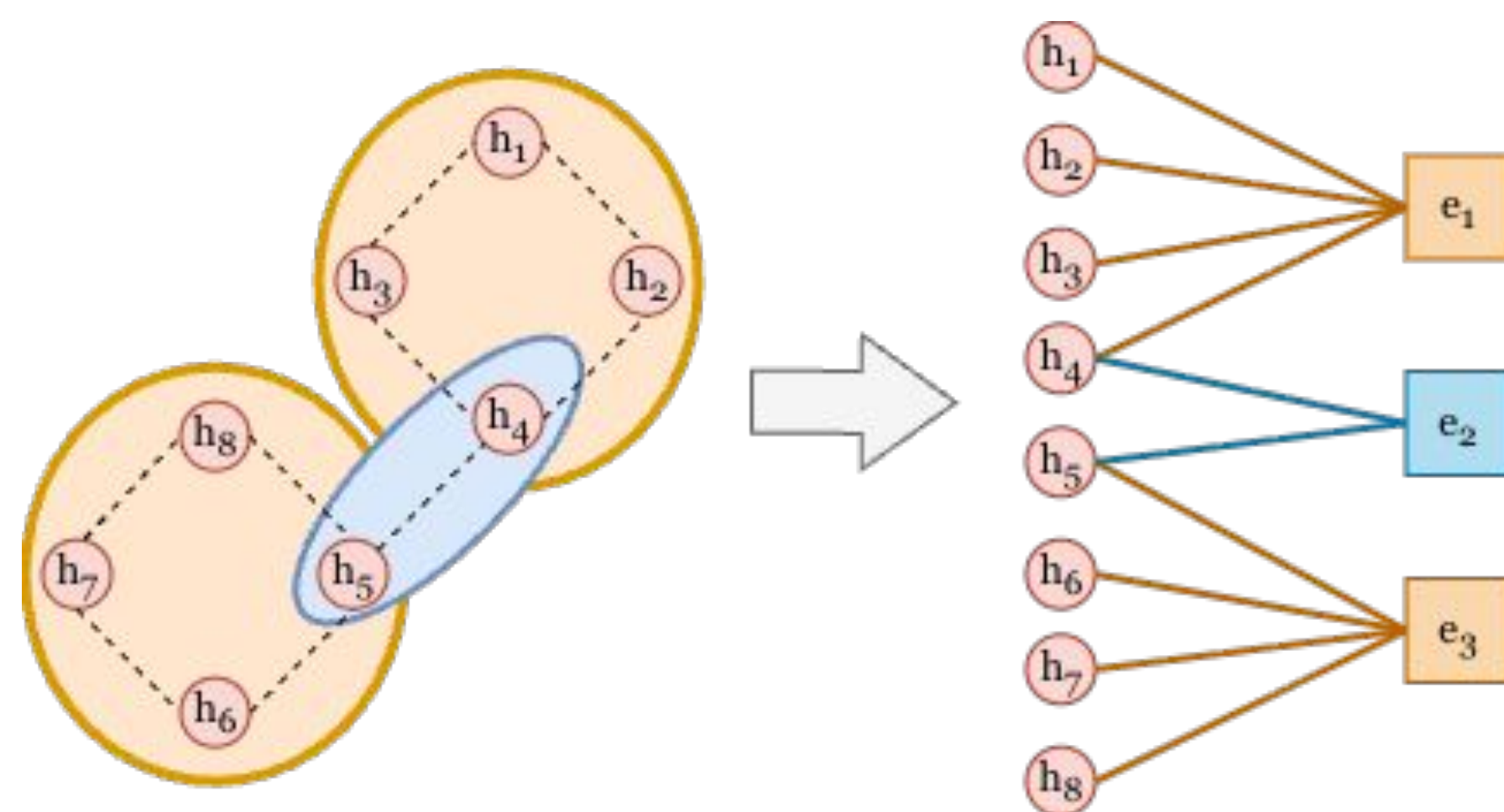
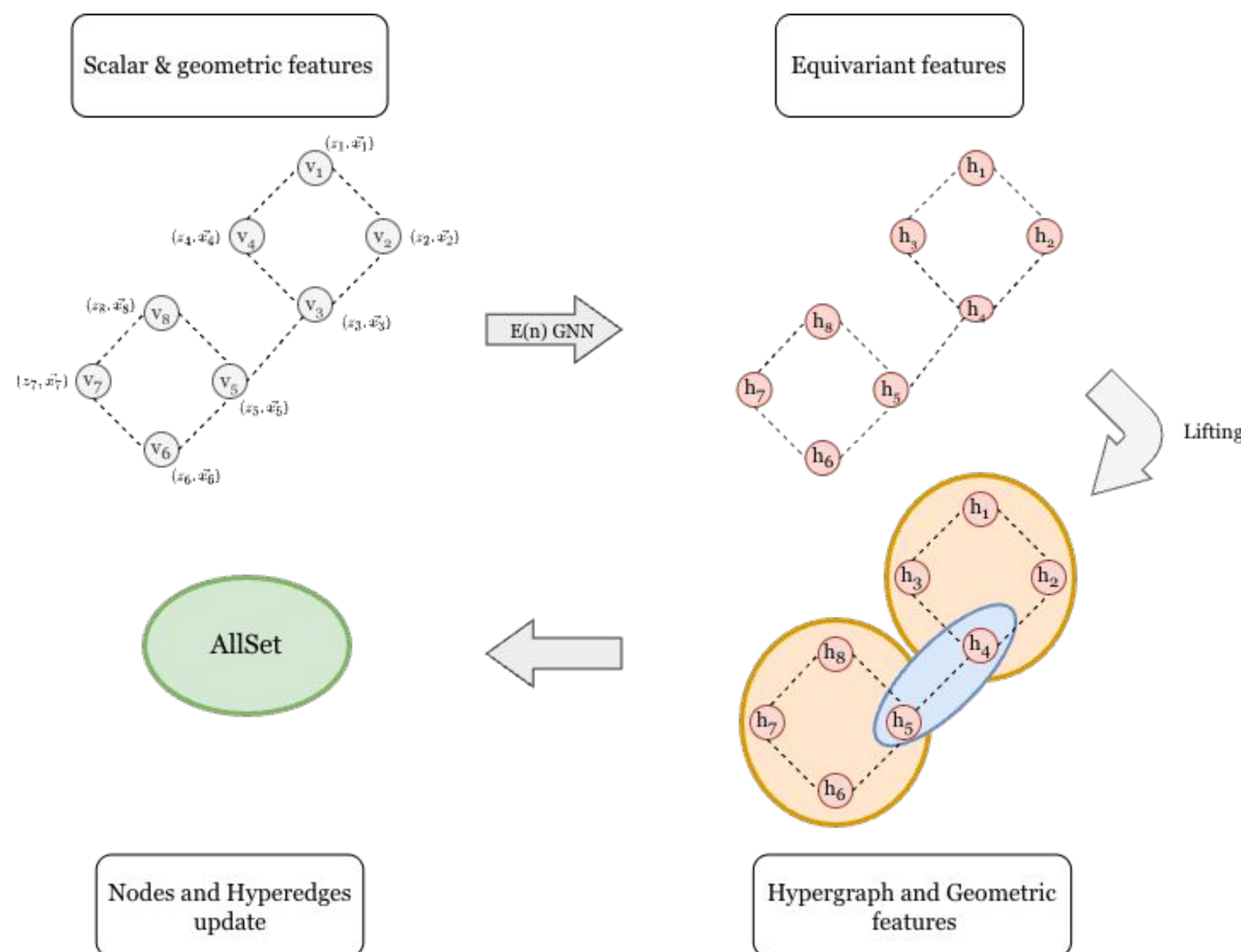


Figure 2: Hypergraph to Bipartite representation

Method

The core idea of EquiHGNN is to combine equivariant geometry with high-order topology.



Node embeddings from 3D coordinates and atomic features are encoded by equivariant models, refined by AllSet for higher-order interactions, and pooled through an MLP to predict energies.

Experiment

Hypergraphs are constructed using RDKit by identifying conjugated π -systems that define chemically meaningful high-order connections.

We evaluate on QM9^[2] (small organic molecules), OPV^[3] (larger organic photovoltaic molecules and polymers), and large-scale benchmarks - PCQM4Mv2^[4] and Molecule3D^[5] to assess scalability and generalization.

Dataset	Graphs	Avg. atoms
QM9 ^[2]	130K molecules	~ 18
OPV ^[3]	90K molecules	~ 200
PCQM4Mv2 ^[4]	3.75M molecules	~ 14.5
Molecule3D ^[5]	3.9M molecules	~24

Table 1: Overview of the datasets

Task	μ	α	ϵ_{HOMO}	ϵ_{LUMO}	$\Delta\epsilon$	$\langle R^2 \rangle$
Units ($\downarrow$)	D	a_0^3	meV	meV	meV	a_0^2
GIN	0.2 $\pm$ 0.003	4.09 $\pm$ 0.04	47.67 $\pm$ 0.4	99.62 $\pm$ 0.9	147.87 $\pm$ 1.3	6279.8 $\pm$ 86.91
GAT	0.65 $\pm$ 0.006	6.17 $\pm$ 0.07	51.56 $\pm$ 0.5	111.24 $\pm$ 0.9	158.26 $\pm$ 1.1	8772.5 $\pm$ 97.43
MHNN	0.67 $\pm$ 0.005	9.29 $\pm$ 0.1	55.38 $\pm$ 0.5	124.23 $\pm$ 1.07	166.6 $\pm$ 1.3	9301.44 $\pm$ 138.369
EGNN-MHNN	0.59 $\pm$ 0.005	2.02 $\pm$ 0.02	44.82 $\pm$ 0.3	92.54 $\pm$ 0.9	<u>140.06 $\pm$ 1.5</u>	<u>4293.09 $\pm$ 66.44</u>
FAFormer-MHNN	<u>0.3 $\pm$ 0.003</u>	4.85 $\pm$ 0.04	<u>26.47 $\pm$ 0.2</u>	51.9 $\pm$ 0.4	73.3 $\pm$ 0.7	2602.89 $\pm$ 25.04
Equiformer-MHNN	0.34 $\pm$ 0.003	<u>2.48 $\pm$ 0.02</u>	25.57 $\pm$ 0.2	<u>67.19 $\pm$ 0.5</u>	230.77 $\pm$ 1.8	102815 $\pm$ 445.433

Table 2: MAE on the QM9 test set

Methods	Molecular				Polymer			
	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	I_{overlap}	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	O_{LUMO}
Units ($\downarrow$)	meV	meV	meV	W/mol	meV	meV	meV	meV
GIN	50.45 $\pm$ 0.9	39.16 $\pm$ 0.5	53.29 $\pm$ 0.8	206.53 $\pm$ 3.6	53.69 $\pm$ 1.0	61.65 $\pm$ 0.9	78.48 $\pm$ 1.5	64.64 $\pm$ 0.6
GAT	55.8 $\pm$ 0.9	32.2 $\pm$ 0.5	46.68 $\pm$ 0.7	204.03 $\pm$ 4.2	47.91 $\pm$ 0.83	58.47 $\pm$ 0.92	71.84 $\pm$ 1.3	56.61 $\pm$ 0.7
MHNN	34.02 $\pm$ 0.4	26.21 $\pm$ 0.4	24.46 $\pm$ 0.3	139.58 $\pm$ 2.3	48.95 $\pm$ 1.1	49.93 $\pm$ 0.8	<u>60.71 $\pm$ 1.1</u>	<u>48.41 $\pm$ 0.7</u>
EGNN-MHNN	<u>28.27 $\pm$ 0.3</u>	20.97 $\pm$ 0.2	<u>20.03 $\pm$ 0.3</u>	99.7 $\pm$ 1.5	<u>45.63 $\pm$ 0.9</u>	66.67 $\pm$ 1.1	69.32 $\pm$ 1.2	67.28 $\pm$ 0.9
FAFormer-MHNN	36.4 $\pm$ 0.6	<u>20.5 $\pm$ 0.2</u>	18.84 $\pm$ 0.3	<u>100.52 $\pm$ 1.5</u>	46.12 $\pm$ 0.9	54.85 $\pm$ 1.0	72.05 $\pm$ 1.4	52.74 $\pm$ 0.8
Equiformer-MHNN	28.12 $\pm$ 0.4	20.24 $\pm$ 0.2	20.59 $\pm$ 0.3	107.346 $\pm$ 1.7	45.42 $\pm$ 0.9	<u>50.08 $\pm$ 0.8</u>	58.17 $\pm$ 1.0	43.6 $\pm$ 0.6

Table 3: MAE on the OPV test set

Model	PCQM4Mv2 gap ($\downarrow$)	Molecule3D gap ($\downarrow$)
GIN	117.65 $\pm$ 0.23	129.61 $\pm$ 0.25
GAT	116.93 $\pm$ 0.21	137.22 $\pm$ 0.36
MHNN	<u>108.11 $\pm$ 0.25</u>	117.55 $\pm$ 0.28
EGNN-MHNN	98.45 $\pm$ 0.20	<u>122.25 $\pm$ 0.24</u>

Table 4: MAE on the PCQM4Mv2 and Molecule3D test set

We compare three approaches: standard 2D graph models (GIN, GAT), hypergraph models using MHNN^[6] (an implementation of AllSet), and equivariant hypergraphs. In EquiHGNN, each variant combines an equivariant model with MHNN: EGNN-MHNN for Euclidean^[7], FAFormer-MHNN for frame^[8], and Equiformer-MHNN for Fourier domains^[9].

Overall, high-order interactions provide modest gains on small molecules but consistently improve performance on large-scale graphs.

References

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- [6] Chen & Schwaller, Molecular Hypergraph Neural Networks, 2024.
- [7] Garcia Satorras et al., E(n) Equivariant Graph Neural Networks, 2021.
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